Demetrius Hernandez

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Bio

I am a second-year Ph.D. student in the Computer Science and Engineering department at the University of Notre Dame, where I am a Sorin Fellow and conduct research on autonomous drones for emergency response scenarios and human-drone interaction under the guidance of Dr. Jane Cleland-Huang. I am also a National Science Foundation (NSF) Computer and Information Science and Engineering Graduate Research Fellow (CSGrad4US) and a Phase 3 Department of Defense (DoD) Science, Mathematics, and Research for Transformation (SMART) Scholar.

Education

The University of Notre Dame — Notre Dame, IN
Ph.D. in Computer Science — *Expected: Dec. 2028*

- Advisor: Dr. Jane Cleland-Huang
- Research Focus: Decision-making and reasoning for autonomous drones in emergency response scenarios

The University of Texas at El Paso (UTEP) — El Paso, TX
B.S. in Computer Science, Minor in Mathematics (Cum Laude) — *Dec. 2022*

- Coursework in Research Methods, Machine Learning, Deep Learning, Data Structures, Data Mining, Database Management, Discrete Mathematics, Probability and Statistics, Matrix Algebra

Technical Skills

Proficient Programming Languages: Python, R, Julia, C++, MATLAB
Machine Learning & AI: TensorFlow, Keras, PyTorch, Scikit-learn, Reinforcement Learning, Deep Learning
Other Tools & Frameworks: Docker, Kubernetes, Git, .NET, Linux

Work Experience

PhD Researcher — Aug. 2024 – PRESENT
The University of Notre Dame — *Notre Dame, IN*

- National Science Foundation (NSF) Research Fellow

Computer Scientist — Feb. 2023 – Aug. 2024
White Sands Missile Range, Counter Drone Team — *White Sands Missile Range, NM*

- Awarded FY 2023 DOD SMART Scholar of the Year Award
- Awarded the Commanders Coin for Excellence by the White Sands Test Center Commander
- Spearheaded automation of analysis processes, conducted research on Graph Theoretic Approaches for Evaluation of Counter Drone Systems, and developed a neural network-based simulation framework, reducing testing costs.

Undergraduate Researcher — Jan. 2022 – Dec. 2022
UTEP Computer Science Department — *El Paso, TX*

- Collaborated with interdisciplinary teams to enhance long-read alignments of DNA/RNA sequences and built neural network architectures, utilizing TensorFlow, to improve DNA assembly through encoding existing minimizer schemes.

Undergraduate Research Assistant — Sep. 2021 – Jan. 2022
UTEP Brain Computation Lab — *El Paso, TX*

- Researched and implemented a comprehensive decision-making model emulating brain circuits, contributing to the lab's objective of integrating big data analysis and behavioral physiology to investigate psychiatric and neurological disorders

Software Engineering Intern — May 2022 – Aug. 2022
White Sands Missile Range, Survivability and Vulnerability Directorate — *White Sands Missile Range, NM*

- Built software to control instrumentation and collect data from equipment such as antennas and spectrum analyzers. Ensured compliance with electromagnetic (EM) requirements for testing weapons systems in EM environments.

Research Experience for Undergraduates Jul. 2021 – Sep. 2021
Oregon State University *Corvallis, OR*

- Utilized machine learning techniques to conduct research in quantitative ecology, analyzing community science data and adapting occupancy modeling frameworks to emerging datasets from the eBird app.

Undergraduate Research Assistant Jul. 2020 - Jul. 2021
UTEP Cyber-Share Center of Excellence *El Paso, TX*

- Analyzed and visualized the Cafeteria Roenbergensis virus, employing high-performance computing resources to contribute to the understanding of its intricate structure composed of approximately 120 million atoms.

Disney College Program Intern Aug. 2019 – Jan. 2020
The Walt Disney Company *Lake Buena Vista, FL*

- Participated in Disney's Ultimate EnginEARing Exploration, Engineering Exploration seminars, and Disney Hackathon, developing transferable skills in problem-solving, teamwork, guest service, and effective communication.

Publications

Ambient Advisory Models: Augmenting Runtime Models into Distributed Reasoning Agents, Hernandez, D., Cleland-Huang, J. *28th International Conference on Model Driven Engineering and Systems.*

Navigating the Black Box: Operational Lenses for AI-Enabled Drone Governance, Hernandez, D., Harris, K., Hernandez, T., Morales, R. *MIT Science Policy Review.*

Graph Theoretic Approaches for Evaluating Counter Drone Systems, Hernandez, D. *17th NATO Operations Research and Analysis Conference Proceedings.*

Aquatic Ecotoxicology: Theoretical Explanation of Empirical Formulas, Hernandez D., Holguin G., Parra F., Sanchez V., Kreinovich V. *Uncertainty, Constraints, and Decision Making. Studies in Systems, Decision and Control.*

Fellowships and Grants

(2025) Notre Dame-IBM Technology Ethics Lab Call for Proposals: Co-author and awarded $38,000 to support research on human-drone collaboration and AI ethics.

(2024) National Science Foundation (NSF) CSGrad4US Fellowship: Awarded to persons in industry with demonstrated potential for doctoral research. $37,000 for 3 years with an additional $16,000 per year for cost-of-education.

(2023) Department of Defense (DoD) Creative Research and Engineering Advancing Technical Equity in STEM Grant: $10,000 award for custom drone development in support of enhancing counter-drone testing capabilities.

(2022) RECOMB Travel Fellowship: RECOMB Travel fellowship

(2021) DOD SMART Scholarship: Full-tuition for undergrad and $25,000/year stipend.

(2018) UTEP Andalusite Award: $3,000 per year for 4 years.

Awards and Honors

(2025) Best Oral Presentation: Graduate Research Symposium, University of Notre Dame
(2025) Finalist: Three Minute Thesis (3MT) Competition, University of Notre Dame
(2025) Honorable Mention: Policy Pitch Competition, Notre Dame Science Policy Initiative
(2024) Congressional Recognition: Received a Certificate of Special Congressional Recognition
(2024) Commanders Coin for Excellence: Awarded by the Commander of White Sands Test Center
(2024) DOD SMART Scholar of the Year: Department of Defense SMART Scholarship Program
(2022) Best Poster Presentation: UTEP Undergraduate Research Symposium
(2021) First Place: Association for Computing Machinery (ACM) Sunrise Hackathon

Service

(2025) Communications Director, Notre Dame Science Policy Initiative
(2025) Graduate Student Board Member, Notre Dame Computer Science and Engineering Department
(2025) Inclusive Excellence and Engagement Committee Member, Notre Dame College of Engineering
(2024-present) Advisory Board Member, UTEP Center for Research in Engineering and Technology Education

Leadership and Service

Mentor Aug. 2021 – PRESENT

Army Educational Outreach Gains in the Education of Mathematics and Science *White Sands Missile Range, NM*

- Mentored high school students in the AEOP GEMS summer program, providing guidance, facilitating workshops, and fostering a positive learning environment, with a commitment to continuing in the role.